

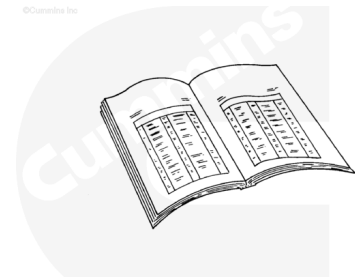
Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

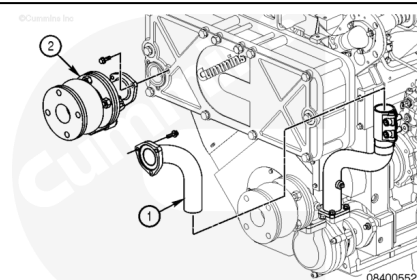
Use caution when draining coolant so that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

- Disconnect the batteries. See equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Remove the coolant vent lines. Refer to Procedure 008-017 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>)

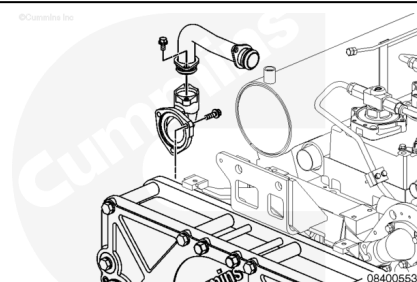
- Remove the aftercooler coolant tube. Refer to Procedure 010-006 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-006.html>)
- Remove the thermostat housing. Refer to Procedure 008-013 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>)
- Remove the expansion tank. Refer to Procedure 008-052 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-052-tr.html>)

Remove

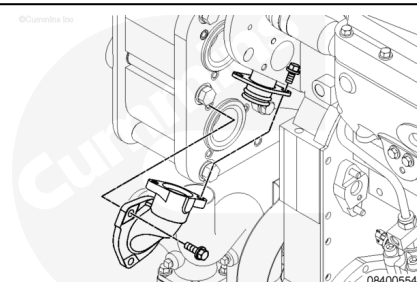
Disconnect **all** the plumbing on the front cover (2) and rear cover (1) of heat exchanger.



Disconnect the engine water inlet.

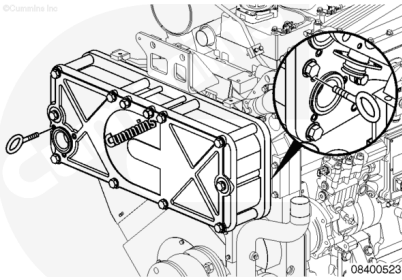


Disconnect the engine water outlet.



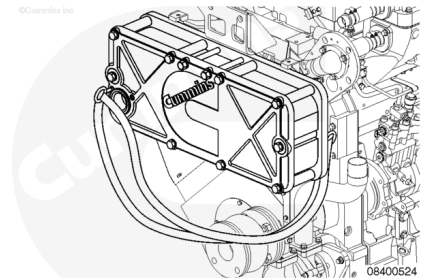
Install a lifting eye into the heat exchanger front plate.

Install a lifting eye into the heat exchanger rear plate.



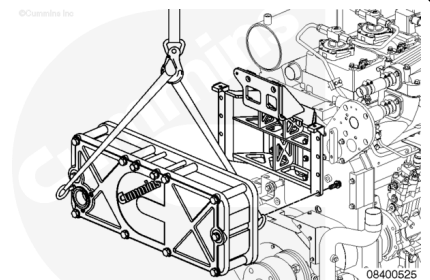
The nylon strap used to remove the heat exchanger **must** have a capacity of 140 kg [300 lb].

Insert a nylon strap through the lifting eyes.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



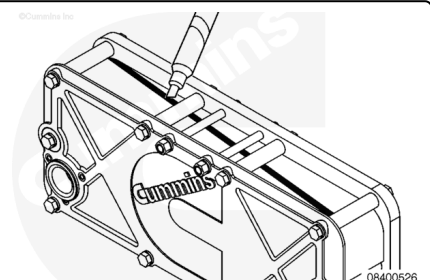
Remove the capscrews from the heat exchanger bracket.

Remove the heat exchanger.

Disassemble

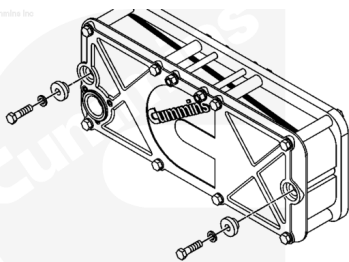
Draw a diagonal line across the top of the plate pack with a paint marker or a wide marker.

The diagonal line will aid in assembly.



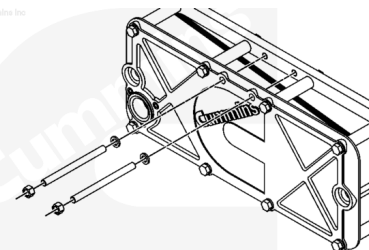
Remove the spacer capscrews.

Remove the inserts that the spacer capscrews were routed through.



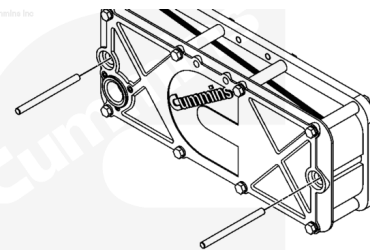
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Remove the service spacers from the front cover.



08400528

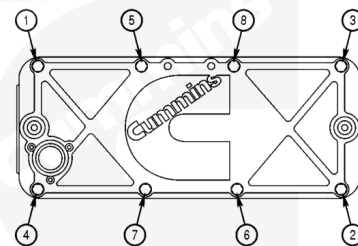
Thread the service spacers into the side alignment spacers until they bottom out.



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The capscrews (1) through (8) **must** be loosened in sequence and in small increments keeping the front cover and back cover within 12 mm [0.5 in] variance from parallel.

Loosen the plate pack clamping capscrews in sequence and in small increments until they can be removed from the assembly.



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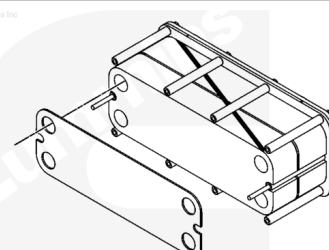
⚠ WARNING ⚠

The plates can have sharp edges. To reduce the possibility of cuts, wear gloves when handling the plates.

Remove the plate clamping capscrews and front cover.

Note : Keep the plates in the order they were removed when disassembling the plates. This will aid in the assembly of the heat exchanger.

Remove the plates from the assembly.

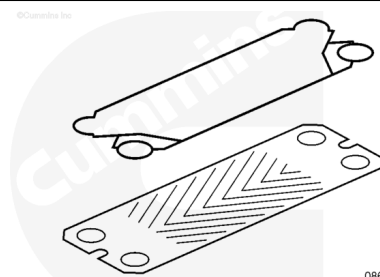


08600574

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



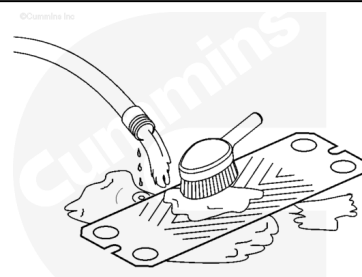
⚠ CAUTION ⚠

Incorrect orientation of the gaskets can cause overheating and sea water to enter the cooling system.

Observe the orientation of the gasket for assembly accuracy.

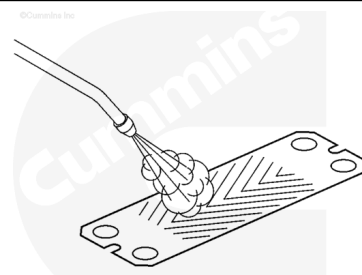
Remove each gasket from the plate one at a time.

Clean the plate with a soft brush and running water.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury



If deposits are present that are difficult to remove, spray the plate with a high pressure water hose.

If brushing and high pressure water does **not** properly clean the plates, evaluate the deposits to determine the type of solvent to be used.

Incrustation

- Calcium carbonate (corrosion products - clean with nitric acid).
- Calcium sulphate (metal oxides - clean with sulfamic acid).
- Silicates (silt - clean with citric acid, alumina - clean with phosphoric acid. Diatomic organisms and their excrement of various colors - clean with complexing agents EDT, A and NTA).

Biological Growth

- Slime - clean with alkaline cleaning agents.
- Bacteria - clean with sodium hydroxide.
- Nematodes - clean with sodium hydroxide.
- Protozoa - cleaning effect can be considerably increased by the addition of small quantities of hypochlorite or agents for the formation of complexes and surfactants.

Concentration maximum - 4 percent.

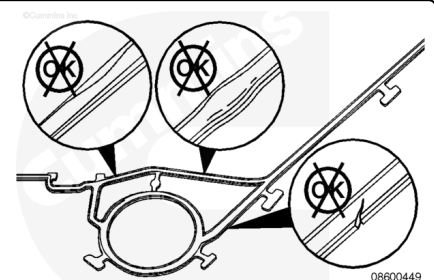
Temperature maximum - 60°C [140°F].

Solvents that can **not** be used:

- Ketones (acetone, methylethylketone, methylisobutylketone)
- Esters (ethylacetate, butylacetate)
- Halogenated hydrocarbons (chloroethene, carbon tetrachloride, freons)
- Aromatics (benzene, toluene).

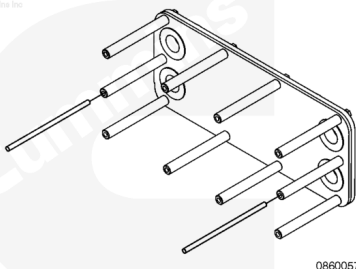
Inspect the gasket for swelling, cracks, and deformation.

Replace the gasket if swollen, cracked or deformed.



Assemble

Insert the service spacers into the back cover side alignment spacers.



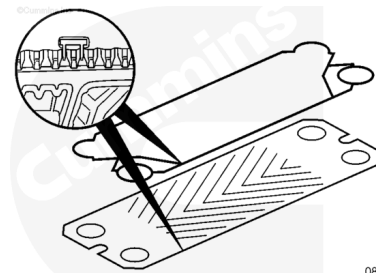
08600575

⚠ WARNING ⚠

The plates can have sharp edges. To reduce the possibility of cuts, wear gloves when handling the plates.

⚠ CAUTION ⚠

Incorrect orientation of the gaskets can cause overheating and sea water to enter the cooling system.



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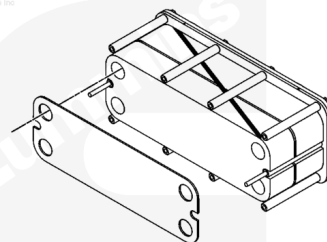
Place the gasket back onto the plates one at a time.

Make sure the tabs are locked into the slots and are orientated correctly.

⚠ WARNING ⚠

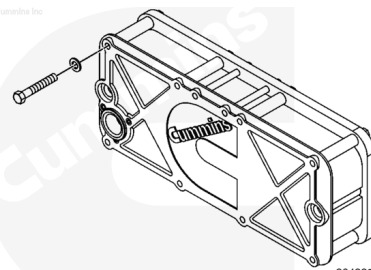
The plates can have sharp edges. To reduce the possibility of cuts, wear gloves when handling the plates.

Insert the cleaned plates against the back cover and in the correct order.



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Install the front cover making sure the aligning shroud groove is on the same side (top) as the alignment groove on the rear cover.



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Install the capscrews (1) through (8) into the assembly.

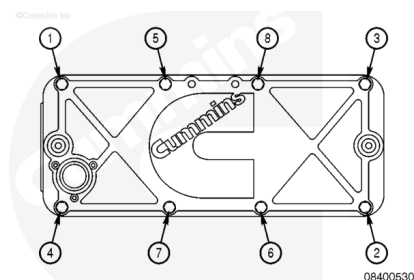
When tightening plate pack clamping capscrews, keep the front cover and the back cover within 12 mm [0.5 in] variance from parallel.

Tighten the capscrews in the sequence shown in the illustration.

Torque Value: 15 n•m [133 in-lb]

Once the front cover is completely against the eight spacers, tighten the plate pack clamping capscrews in the sequence in the illustration.

Torque Value: 135 n•m [100 ft-lb]

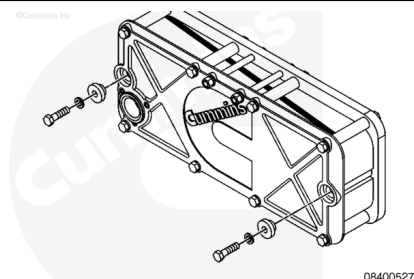


Remove the service spacers.

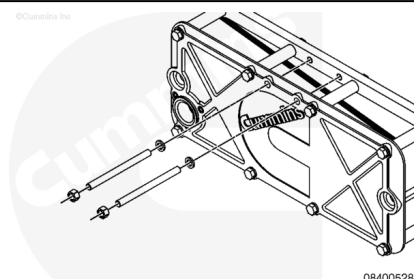
Install the inserts and side alignment capscrews into the two locations.

Tighten the capscrews.

Torque Value: 88 n•m [65 ft-lb]



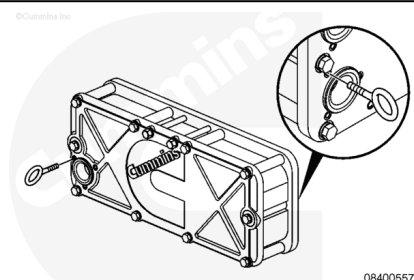
Install the service spacers into the two locations in the illustration.



Install

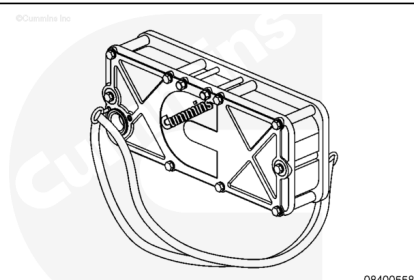
Install a lifting eye into the heat exchanger front plate.

Install a lifting eye into the heat exchanger rear plate.



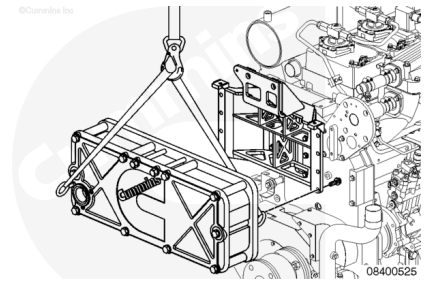
The nylon strap used to install the heat exchanger **must** have a capacity of 140 kg [300 lb].

Insert a nylon strap through the lifting eyes.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



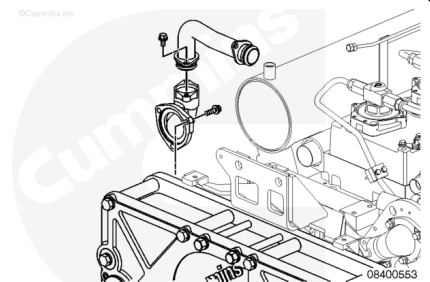
Place the heat exchanger onto the bracket.

Insert the bracket capscrews that attach the bracket to the heat exchanger.

Torque Value: 45 n•m [33 ft-lb]

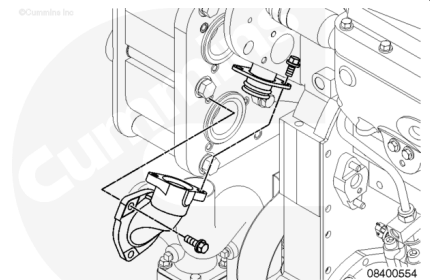
Remove the nylon strap.

Connect the engine water inlet.

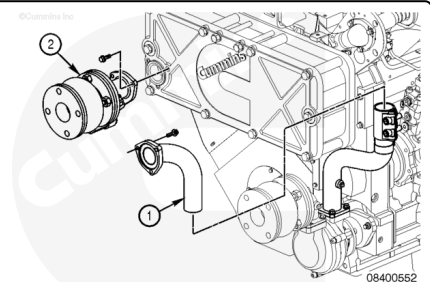


Connect the engine water outlet.

Torque Value: 45 n•m [33 ft-lb]



Install the sea water outlet (2) and the sea water inlet (1) on the heat exchanger.



Delete and Reset Alarm Codes

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ CAUTION ⚠

Use caution when draining coolant so that the coolant is not spilled or drained into the bilge area. Do not pump coolant overboard. If the coolant is not reused, it must be disposed of in accordance with local environmental regulations.

- Install the expansion tank. Refer to Procedure 008-052 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-052-tr.html>)
- Install the thermostat housing. Refer to Procedure 008-013 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-013-tr.html>)
- Install the aftercooler coolant tube. Refer to Procedure 010-006 in Section (</qs3/pubsys2/xml/en/procedures/20/20-010-006.html>)
- Install the coolant vent lines. Refer to Procedure 008-017 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-017.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)
- Connect the batteries. See equipment manufacturer service information.